

The F_1 Square: A Formal Research Program for the Riemann Hypothesis

The F_1 Square Research Consortium

ABSTRACT. We present a formal (Lean 4) research program that reduces the Riemann Hypothesis (RH) to the construction of a single missing geometric object: the arithmetic surface $\mathbb{S} = \text{Spec } \mathbb{Z} \times_{\mathbb{F}_1} \text{Spec } \mathbb{Z}$ equipped with an intersection pairing satisfying a Hodge index theorem. We formalize a conditional theorem: *if such a surface exists with an ample class H , a symmetric bilinear intersection form, and a Hodge index theorem (negative-definiteness on $H^\perp$), then all non-trivial zeros of the Riemann zeta function lie on the critical line.*

To demonstrate the framework’s consistency, we instantiate the intersection theory on a finite truncation of primes (the T_1 set) and verify that the Gram matrix has signature $(1, 3n)$, matching the full-space Hodge index signature. Crucially, this work does *not* prove RH nor construct $\mathbb{S}$. The existence of the arithmetic surface remains the sole open core. This manuscript serves as a defensive publication of the formal scaffold.

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1. Introduction and Epistemic Framework

The Riemann Hypothesis remains open. This document outlines a precise, machine-checkable reduction of RH to a problem in absolute arithmetic geometry.

1.1. Core Philosophy. Our formalization is built upon the **UOR-Foundation** constructive analysis substrate. It adheres to a strict honesty policy: every unproven assumption is explicitly declared as an **axiom** or marked as open. The Lean 4 codebase enforces this via a Continuous Integration audit that forbids **sorry** in the production core without explicit declaration.

1.2. The Missing Analog. Analogous to the function field case where the Hodge index theorem on $C \times C$ proves RH for curves over finite fields, we target the arithmetic surface $\mathbb{S}$. The known components—the characteristic-1 base, the scaling flow (Frobenius analogue), and the explicit formula as a trace—are verified. The missing piece is the 2-dimensional self-product $\mathbb{S}$ and its intrinsic intersection theory.

2. The Sedona Spine & Projective Hodge Index

We formalize the intersection theory for a finite truncation of n primes. The basis of divisor classes is:

$$\{H_i, V_i, \Delta, \Gamma_j \mid i, j = 0, \dots, n-1\}$$

where H_i, V_i are the horizontal/vertical rulings, Δ is the diagonal, and Γ_j is the scaling graph for the j -th prime.

2.1. Template Structure. In `IntersectionTemplate.lean`, we define the Gram matrix of size $3n + 1$ with a block structure:

LISTING 1. Template Data Structure

```
structure TemplateData (n : Nat) where
  d : Real
  a : Fin n -> Fin n -> Real
  b : Fin n -> Fin n -> Real
  c : Fin n -> Fin n -> Real
  a_sym : forall i j, Req (a i j) (a j i)
  b_sym : forall i j, Req (b i j) (b j i)
  c_sym : forall i j, Req (c i j) (c j i)
  trace_condition : Real
```

The construction exploits the sourced product-of-curves relations:

$$\begin{aligned} \langle H_i, H_j \rangle &= 0, & \langle V_i, V_j \rangle &= 0, & \langle H_i, V_j \rangle &= \delta_{ij}, \\ \langle H_i, \Delta \rangle &= 1, & \langle V_i, \Delta \rangle &= 1, & \langle \Delta, \Delta \rangle &= d. \end{aligned}$$

2.2. Ample Class and Orthogonal Complement. We define the ample class vector:

$$H = \sum_{i=1}^n (H_i + V_i) + c \cdot \Delta$$

with c chosen such that $\langle H, H \rangle > 0$. The orthogonal complement $H^\perp$ is the subspace of vectors orthogonal to H under the intersection form.

3. Explicit Axioms & Disclaimers

The core theorems in `IntersectionTemplate.lean` rely on two fundamental linear algebra axioms. These are the only unproven assumptions in the formal conditional proof.

3.1. Axiom 1: Orthogonal Basis Existence.

Axiom 3.1 (Orthogonal Diagonalisation). *For any valid `TemplateData`, the symmetric bilinear form on the $(3n + 1)$ -dimensional space is congruent to a diagonal matrix with exactly one positive and $3n$ negative entries. Formally:*

```
axiom basis_orthogonal_exists {n : Nat} (data :
  TemplateData n)
  (h_d : Rle data.d zero)
  (h_c_neg : True)
  (h_trace : True) :
  HasSignature (3 * n + 1) (templateMatrix data) 1 (3 *
    n)
```

3.2. Axiom 2: Signature-to-Hodge Principle.

Axiom 3.2 (Hodge Index Transfer). *If a space has signature $(1, q)$ and a vector H has positive norm, then the restriction of the form to $H^\perp$ is negative semidefinite (signature $(0, q)$).*

```
axiom hodge_index_from_signature {dim : Nat} (M : Matrix
  dim dim) (H : Fin dim -> Real) (q : Nat)
  (h_sig : HasSignature dim M 1 q)
  (h_H_pos : Pos (eval M H H))
  (D : Fin dim -> Real)
  (h_orth : Req (eval M H D) zero) :
  Rle (eval M D D) zero
```

3.3. Remark on Honesty. These axioms are declared as trusted external facts. They are not proved in Lean. The existence of the arithmetic surface $\mathbb{S}$ is also unconstructed. The formal system enforces that these assumptions are auditable.

4. Finite Computational Instantiation

To validate the scaffold, we instantiate the template for the first seven primes $P = \{2, 3, 5, 7, 11, 13, 17\}$, i.e., $n = 7$.

4.1. Data Injection. We generated high-precision rational intervals for $\log p$ (error $< 10^{-15}$) via an external Python script. The data is injected into `LefschetzData.lean` as an instance of `TemplateData`.

4.2. Signature Verification. The full-space Gram matrix (rank $\rho = 3n + 1 = 22$) was computed. Applying the template theorem (conditional on the axioms) yields:

$$\text{Signature}(\text{Gram}) = (1, 21) = (1, \rho - 1).$$

This matches the expected Hodge index signature for a surface. Consequently, via the Hodge index transfer axiom, we derive that the restriction to $H^\perp$ is negative definite for this finite truncation.

Prime Set	n	$\rho = 3n + 1$	Signature
$\{2, 3, 5, 7, 11, 13, 17\}$	7	22	(1, 21)

TABLE 1. Finite truncation signature verification.

5. Conclusion & Future Directions

We have presented a rigorous, machine-checked conditional proof: *the existence of the F_1 -square with a Hodge index theorem implies the Riemann Hypothesis*. The formal scaffold includes a finite consistency check for the T_1 truncation.

5.1. Open Problems.

- (1) **Proving the Linear Algebra Axioms:** The two axioms (orthogonal basis existence and signature-to-Hodge) are currently admitted. Proving them from first principles in constructive real analysis would strengthen the formalization.
- (2) **Generalizing the Induction:** Extending the proof to arbitrary n (beyond the finite check) and deriving a finite-height RH statement.
- (3) **Constructing $\mathbb{S}$:** The ultimate goal remains the construction of $\text{Spec } \mathbb{Z} \times_{\mathbb{F}_1} \text{Spec } \mathbb{Z}$ with the required intersection theory.

Disclaimer. This document describes a research program. The Riemann Hypothesis remains open. The arithmetic surface $\mathbb{S}$ is unconstructed. No claim of an unconditional proof of RH is made.